Team Details

55% of the teams that shipped voice AI abandoned it, top reason first-turn understanding, and Indian field speech recognition errs over 50%. We constrain the decoder, then run it offline on a $5 chip, in any Indian language.

AI & Machine Learning

RISC

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ABOUT THE PROJECT

VĀKYA: a $5 device that hears Indian speech and performs a verified action, offline, forever.

One core. Many instruction sets. The task compiles to a grammar; the grammar IS the instruction set, so nothing task-specific is baked into the weights.

THE FAILURE, MEASURED

- 55% of the 455 teams that built voice agents abandoned the project. Top reason: first-turn understanding.
- 95% of users report frustration with voice AI at some point, and a third still prefer a human.
- Drive-thru voice AI: 85% accuracy. Humans deliver 89-92%. IVR abandonment 38%, and 67% of calls escalate anyway.
- In Indian fields: 50%+ speech-to-text error from dialect, slang and noise. (Farm Vaidya / IFPRI, 2026)
- **It fails silently, and it fails hardest on the people with the least power.**

Every one of those systems answers in an OPEN vocabulary. The failure lives in the gap between what was heard and what was executed.

THE MOVE

Move the constraint from the model to the decoder.

- A byte-level grammar compiled from the target's own schemas constrains every decoded token, so the action is well-formed by construction, in any script.
- **A garbled utterance can only produce a valid action or a refusal. Never a wrong one.**
- Guardrail, audit trail and refusal stop being aspirations and become architecture.

- The grammar becomes the localisation layer: it carries date formats, numeral systems, currency units and enum members, so the model never has to learn them.

THE SYSTEM

One binary. Three hardware tiers. Zero network bytes. Zero marginal cost.

- Tier A $5 MCU, wake word plus a grammar-constrained decoder
- Tier B sub-$200 phone or RPi 5, full multilingual tool calling
- Tier C cloud, only when confidence falls below the threshold

- **One box, any task. Re-point it by recompiling the grammar, not by retraining the model.**

Grounded entirely on open work: Cactus Needle 3 (Apache 2.0), Espressif ESP-SR, Oído.

TECHNICAL APPROACH

MIC
WakeNet wake word, 98% at 1 m, one false trigger per 12 h

▼

SPEECH TO TEXT
Whistle 16.9 MB, 7 languages, CPU only | Oído 13M int8, 3.70% WER on a $5 MCU

▼

ROUTER
script + code-mix + confidence → a language simplex, not a classifier

▼

INDIC SUBNETWORK
4 / 8 / 16 / 20 layers · 29-121M params · 8-29 MB binary

▼

GRAMMAR
byte-level, compiled from the task's schemas → well-formed by construction

▼

OUTCOME
valid action | refusal | cloud handoff → executor + audit log

BUILD github.com/harshgounder/vakya **LIVE** harshgounder.github.io/vakya **DECK** the 6-slide PDF is the same file

THREE TIERS, ONE BINARY

TIER			
A	ESP32-S3, $5	13M int8 ASR + wake word + grammar decoder	8 MB PSRAM, mW, weeks on battery
B	sub-$200 Android / RPi 5	Whistle 16.9 MB + Needle 3, 8-29 MB	500 tok/s, 28-45 MB RAM
C	cloud, only if confidence < threshold	frontier model	metered $0.08-0.20 per minute

Same core, same grammar compiler, three power envelopes. Tier A does the wake word and the constrained decode; tier B does open-ended reasoning; tier C is the safety valve.

THE SUBSTRATE, IN REAL NUMBERS

- 2.125 bits per weight, trained in from pretraining, never post-hoc
- 29 MB binary · 29-121M params · 8-29 MB · 6000 tok/s prefill, 1200 tok/s decode
- 3.70% WER on a $5 microcontroller with no neural accelerator
- a 2-layer subnetwork, fine-tuned on one tool set, runs on devices far smaller than the full model needs

THE THREE CONTRIBUTIONS

- 1. Per-language subnetwork lattice: Tamil pays for 4 layers, code-mixed Hindi pays for 16. One download.
- 2. Grammar as instruction set: one core, one compiled dialect per app, nothing baked into the weights.
- 3. Refusal as a first-class output, calibrated per script, in the user's own register.

FEASIBILITY AND VIABILITY

WHAT IS ALREADY PROVEN

Only the language axis is new. Every hard piece exists, in the open, today.

- Per-task subnetwork fine-tuning lifts every subnetwork 18-36 points; from 4 layers up, a tuned subnetwork passes DeepSeek V4 Flash at 29M params. (Cactus, 2026)
- Open-vocabulary ASR at 3.70% WER on a $5 MCU with no neural accelerator. (Oído, arXiv 2610.13447, 9 Oct 2026)
- Wake word, 200-command recogniser and TTS already run on an ESP32-S3. (Espressif ESP-SR)
- 30,000 hours of real farm audio across 10 Indian languages and 5,000 vernacular terms already collected. (Farm Vaidya)

THE SCALABILITY LINE

At 10x the tool set breaks before the model does.

- Fix: pre-compile and cache the grammar per tool set, shard the tool set by domain, and ship the grammar with the app. The model is the cheap part.

THE UNIT COST

₹0 per action. 0 network bytes. 29 MB of RAM.

- Against a metered API call at $0.08-0.20 per minute, a 38% IVR abandonment rate, and a data-centre round trip for every question.

SIX RISKS, SIX ANSWERS

RISK

1. **No Indic tool-call corpus exists.**
   Synthesise over IndicGenBench and Indic-QA with human verification, the Indic-QA method exactly, and publish it as Indic-ToolCall-Indic.
2. **English-first pretraining, worst on low-resource languages.**
   Per-language subnetworks instead of one Indic model. Translate-Test is the baseline we have to beat, not our method.
3. **Code-mixed speech, field noise, dialect.**
   The grammar bounds the action space, so noise yields a valid action or a refusal, never a malformed one. This is the structural payoff of constraining at the decoder.
4. **Thirteen scripts inside a 29 MB budget.**
   The lattice. Script-clustered vocabularies with shared subword units, and a per-script subnetwork. Worst case falls back to a 20-layer code-mixed subnetwork.
5. **There is no benchmark to be judged on.**
   Propose Indic-ToolCall-Indic as a BFCL and MLCL extension, and publish the set with the model. A published benchmark is a moat.
6. **The buyer is not the user.**
   True, and we name it. The payer is a state programme, an FPO, an NGO or a device maker. At ₹0 marginal cost, a state programme can put 100,000 devices in the field for the cost of the devices.

IMPACT AND BENEFITS

140M
farmers addressable in India

1.86 lakh
Anganwadi centres, 540 crore visits

122+
languages spoken in India

₹0
marginal cost per AI action

EVERYONE HAS ONE PIECE. NOBODY HAS THE LAST MILE.

WHO

Bhashini, Government of India
$600M budget, LLMs for 100 Indian languages. Cloud scale.

Trilateral AI Hub (India, Italy, Africa)
Offline voice for agriculture and fintech, but powered by European supercompute.

FarmerChat / Digital Green
1M+ users, 30,000 h of farm audio, 16-23% WER reduction. Cloud.

Farm Vaidya, Farm Vighnaharta
5,000+ crop Q&A across 100+ crops, Telugu. Cloud, and field WER above 50%.

BharatSahayak, Yuverse, Bharosa AI
Voice-first, 3,400+ schemes, offline on 1 GB RAM. Cloud or app layer.

Cactus, Oído, Espressif
The on-device substrate, at $5 to $150, in English.

THE GAP: offline + $5 + grammar-safe + Indic + zero marginal cost. Nobody has built it, and every hard component to build it is already open.

THE THREE BUCKETS

SOCIAL
Voice is the only interface for the roughly 250 million adults who cannot read a screen. It also removes the human intermediary who today translates the question and fills the form.

ECONOMIC
The marginal cost of an AI action collapses from a metered API call to a local forward pass. No vendor lock, no per-seat pricing, no data leaving the device.

ENVIRONMENTAL
Zero network bytes per action, at 1.4 billion people. Every avoided round trip is avoided data-centre energy.

FIRST ADOPTER
One district health or agricultural app. One sub-$200 Android. One 2G-only taluka. Measured by actions completed per device per day with zero network dependence.

THE COMPILER TEST
Farm advisory, scheme eligibility, price and weather, appointment booking, status check, record lookup. All enumerable, all machine-callable, all safety-critical where a wrong action destroys a crop.

RESEARCH AND REFERENCES

THE FAILURE WE ATTACK

arXiv 2601.05366, Lost in Execution: On the Multilingual Robustness of Tool Calling in LLMs. MLCL extends BFCL with Chinese, Hindi and Igbo.

Semantically appropriate yet operationally invalid. Parameter values must follow execution conventions such as English string identifiers.

Parameter value language mismatch. Failures arise at the language-execution boundary, not from intent misunderstanding. Silent non-executable calls.

arXiv 2404.16816, IndicGenBench: 29 Indic languages, 13 scripts, 4 families, 1.4 billion people.

arXiv 2407.13522 and Findings NAACL 2025, Indic-QA: 11 languages, subpar on low-resource, Translate-Test beats direct multilingual inference.

THE SUBSTRATE WE STAND ON

Cactus Compute, Needle 3 and Cactus engine. Apache 2.0. github.com/cactus-compute/needle, github.com/cactus-compute/cactus, HF Cactus-Compute/needle3.

cactuscompute.com/blog: needle, needle-2, whistle, hybrid, cact-format, finetuning-needle, porting-needle. Includes a published porting oracle to test a runtime against.

arXiv 2607.18363, Simple Attention Networks.

Oído, arXiv 2610.13447, 9 Oct 2026. Open-vocabulary speech recognition on a $5 microcontroller without a neural accelerator.

Espressif ESP-SR: WakeNet wake word, MultiNet with up to 200 commands, and on-chip TTS.

DroidCall, Findings EMNLP 2025. google/mobile-actions, 9,654 rows. Salesforce xlam-function-calling-60k.

THE DEPLOYMENT PRECEDENT

NCMS 2026 ALC exhibit directory. Edlore AI-Server: TRL-6, no cloud connectivity, voice-assisted, peer-to-peer sync, NIST-aligned, in disconnected and contested environments.

NCMS, Artificial Intelligence to Improve Supply Chain Resilience Throughout American Manufacturing, 2024. NCMS Technology Brief on hybrid AI, Oct 2026.

FarmerChat, Digital Green, with a 60 Decibels study: 1M+ users, 60% acted on the advice, 83% said information was easier to access, 91% greater confidence.

Farm Vaidya, IFPRI, Apr 2026: 50%+ field STT error, 30,000 h across 10 Indian languages, target under 3% WER, 5,000+ vernacular terms collected.

BharatGen and IIIT Hyderabad: 13K questions across 11 low-resource Indic languages, best model near 58%.

Bhashini: $600M, 100 Indian languages. Trilateral AI Hub: India, Italy and Africa.

OURS: Indic-ToolCall-Indic, a published benchmark extension · the per-language subnetwork lattice · the grammar-as-instruction-set compiler · the per-script calibration and refusal protocol · the ablation plan

BUILD github.com/harshgounder/vakya LIVE harshgounder.github.io/vakya DECK the 6-slide PDF, rebuilt from the official template by build/build_deck.py